PHYC10009 Physics Foundations

Electromagnetic Induction
Week 9 Lecture 2

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School of Physics
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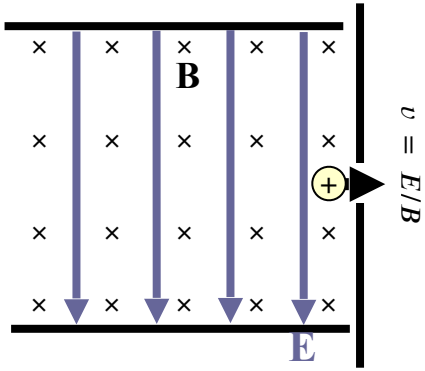
Previous lecture

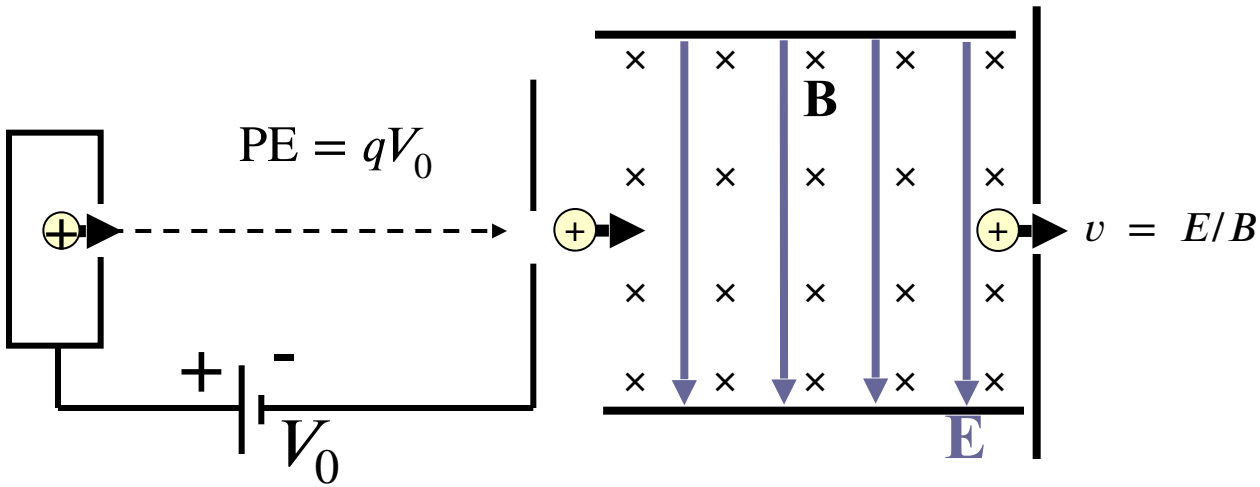
- Force on moving charge in magnetic field
  - $F = qvB\sin\theta$
  - right hand rule with thumb and two fingers

- Wien velocity filter
  - Crossed E and B fields: $v = \dfrac{E}{B}$
  - F_E and F_B equal magnitude, opposite directions



Mass spectrometer

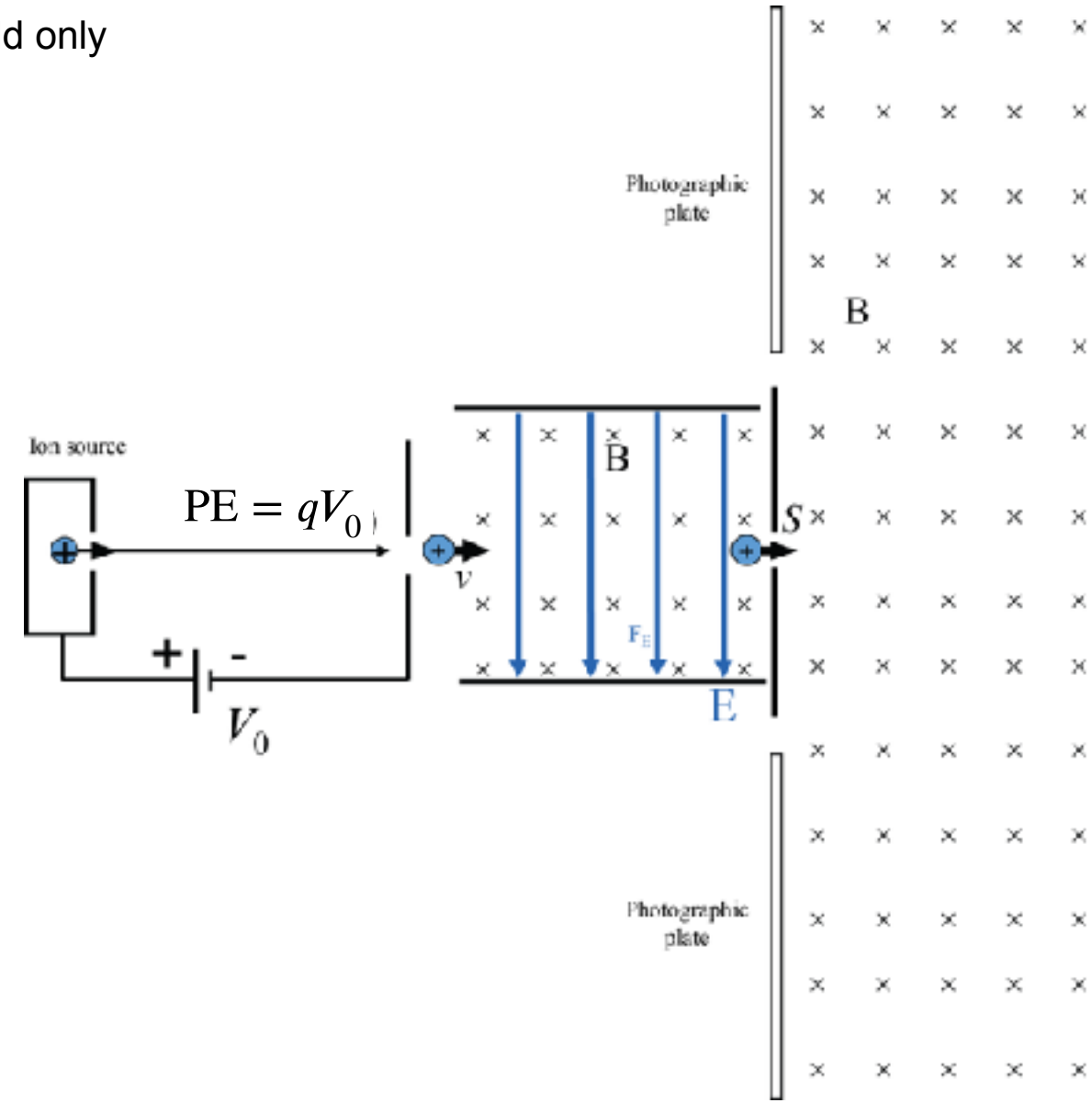
- Charged particles accelerated by a constant V_0
- Energy $\mathrm{PE} = qEd = qV_0 = \frac{1}{2}mv^2$ so $v = \sqrt{\frac{2qV_0}{m}} \propto \sqrt{\frac{q}{m}}$



- Put in unknown materials, tune E/B until detected (usually adjust E)
- E/B gives v hence q/m hence m
- But difficult: $m \propto \left(\frac{B}{E}\right)^2$ (non linear)

20.11 Mass spectrometer (1919)

- After velocity selector: mag field only



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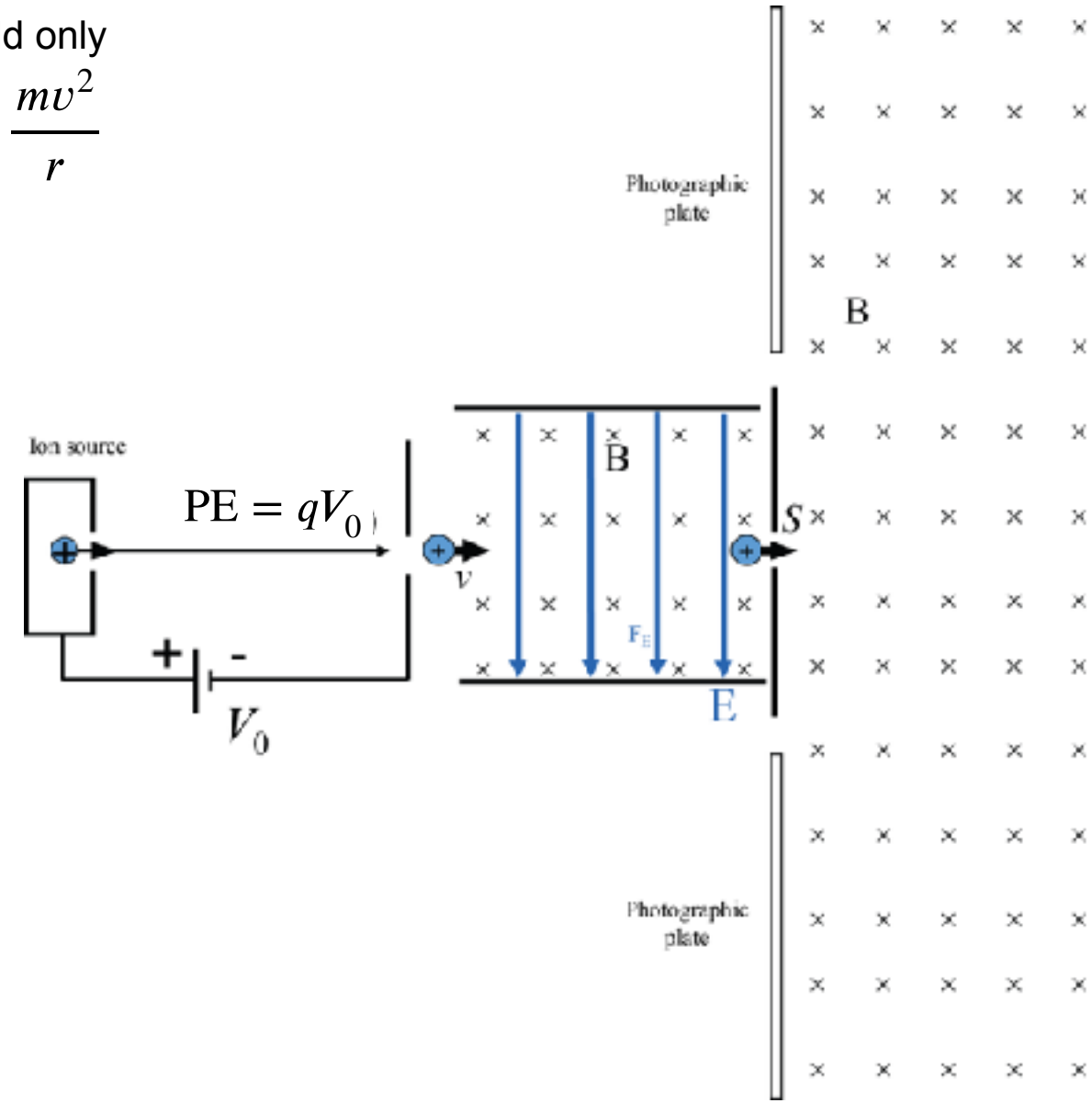

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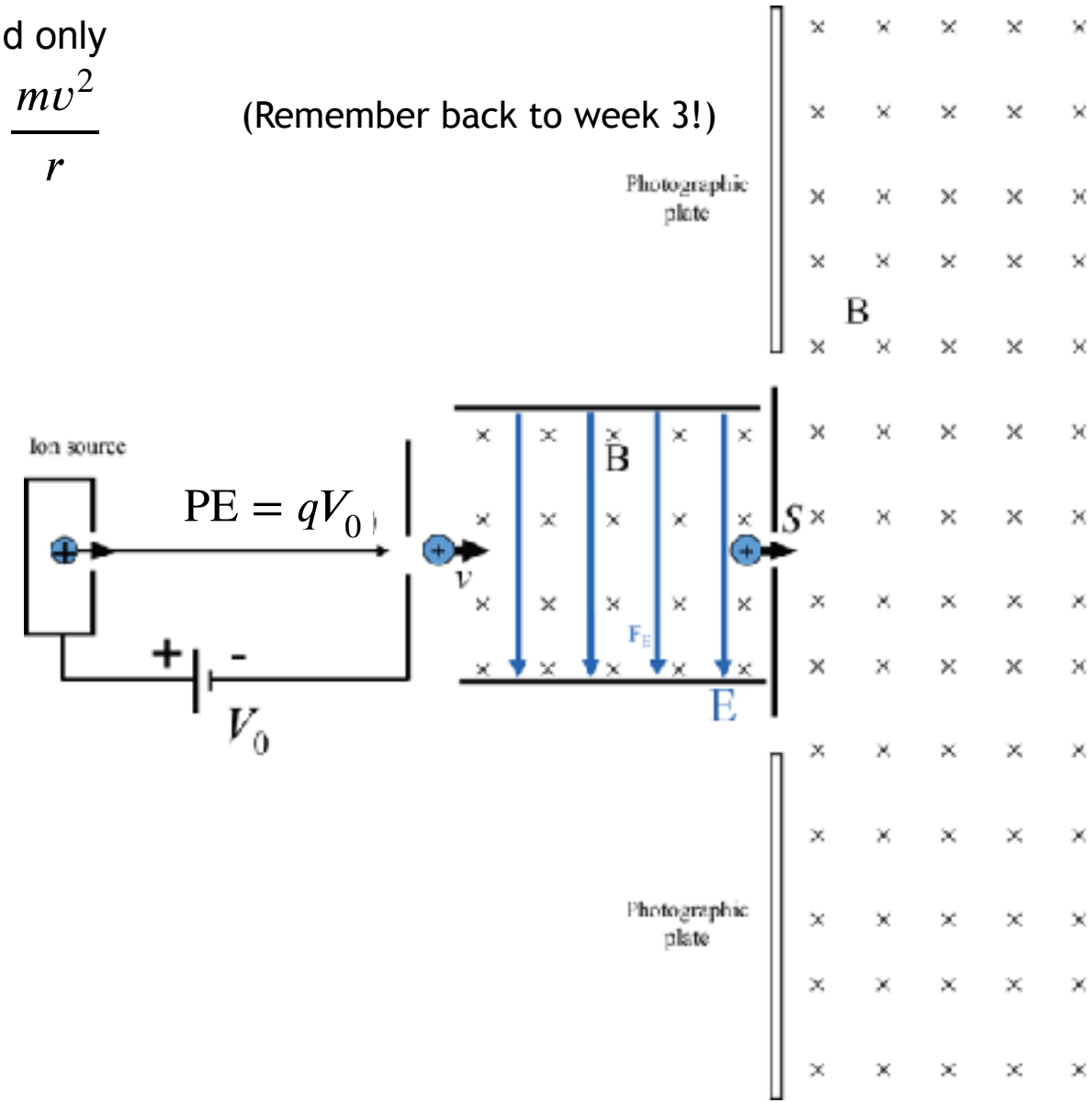

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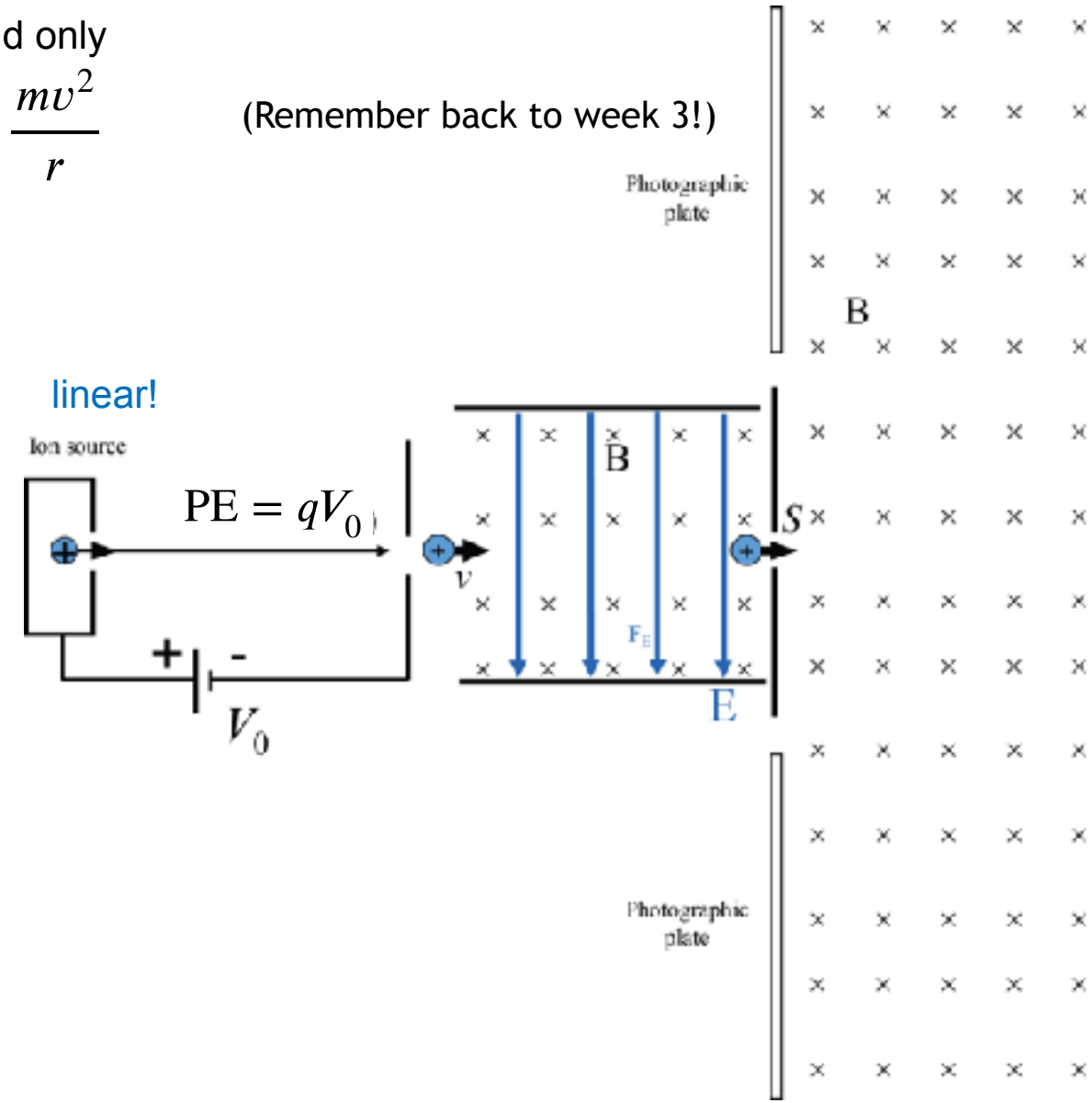
- Circular path: $r = \dfrac{v}{qB}\, m$
- Distance along detector $\propto m$ linear!



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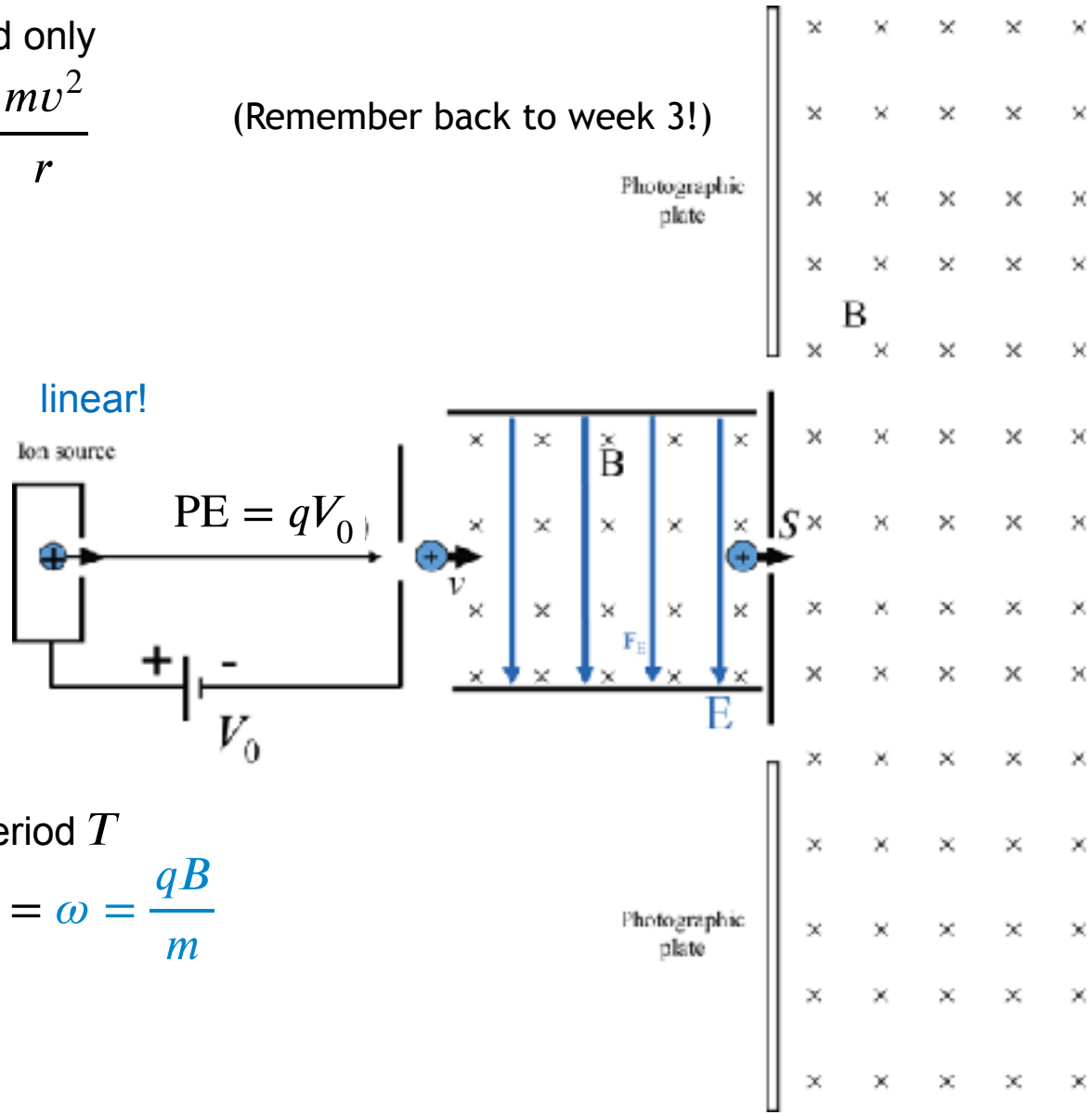

- Orbits: circumference $2\pi r$ in period T

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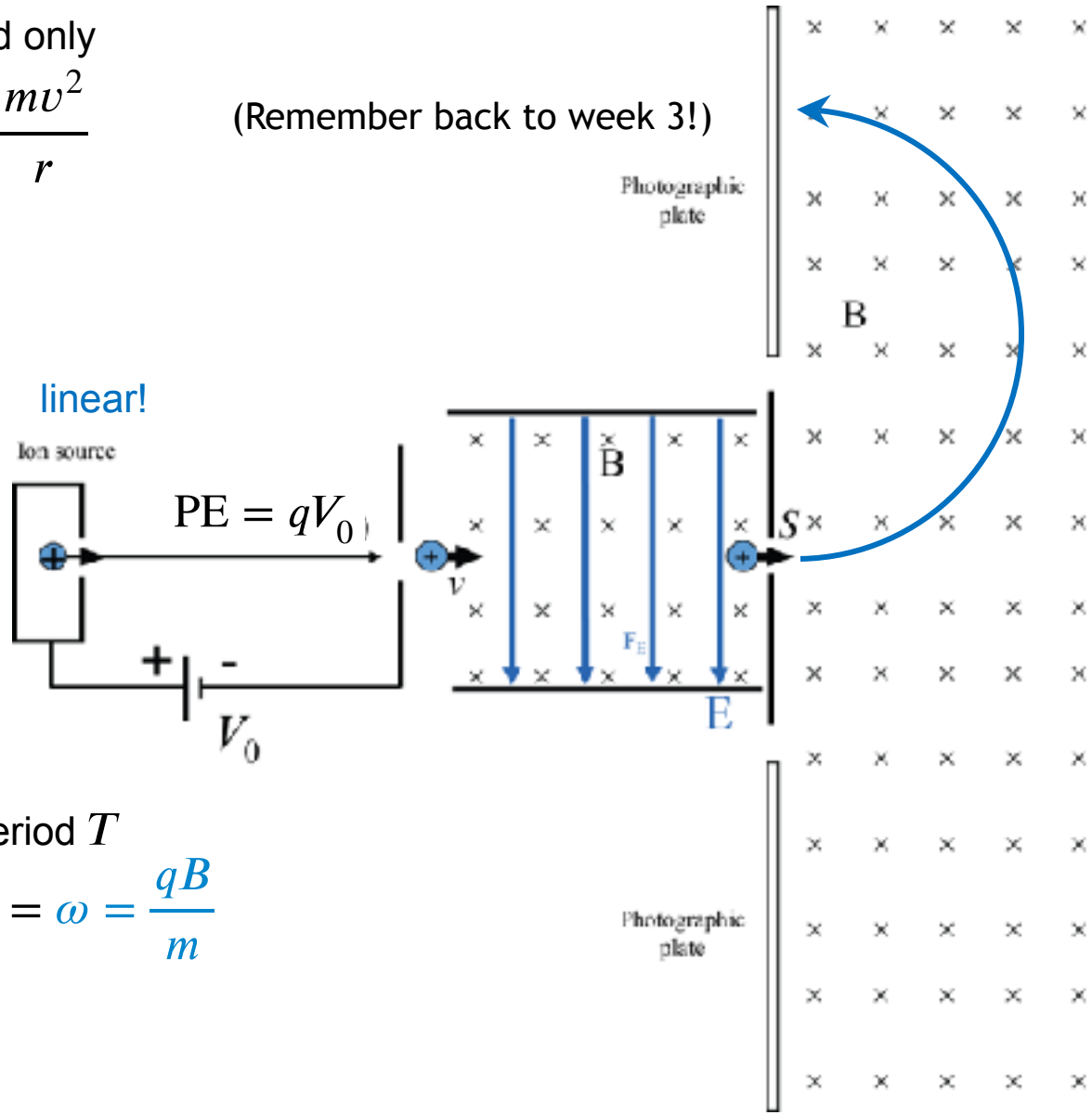

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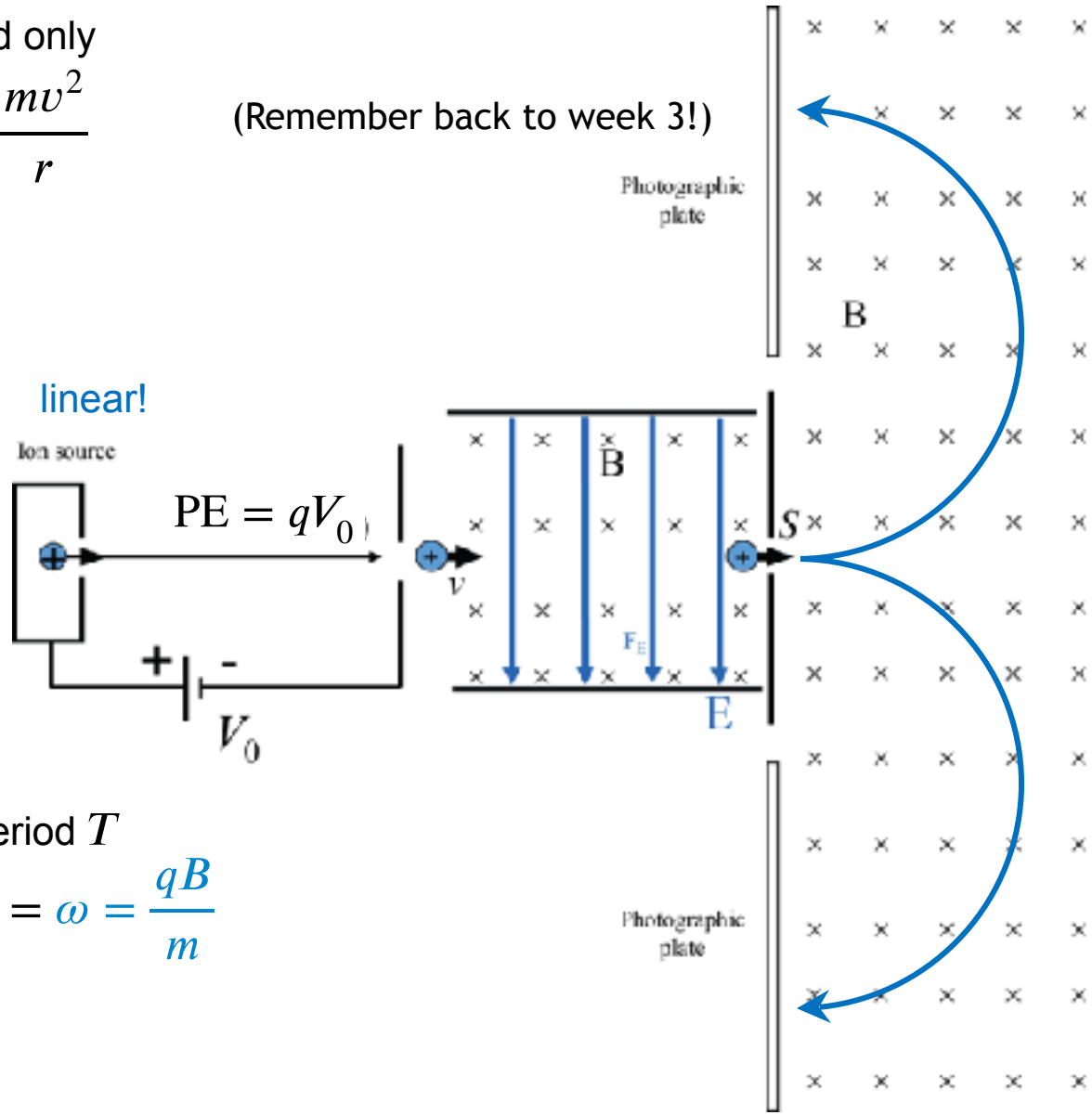

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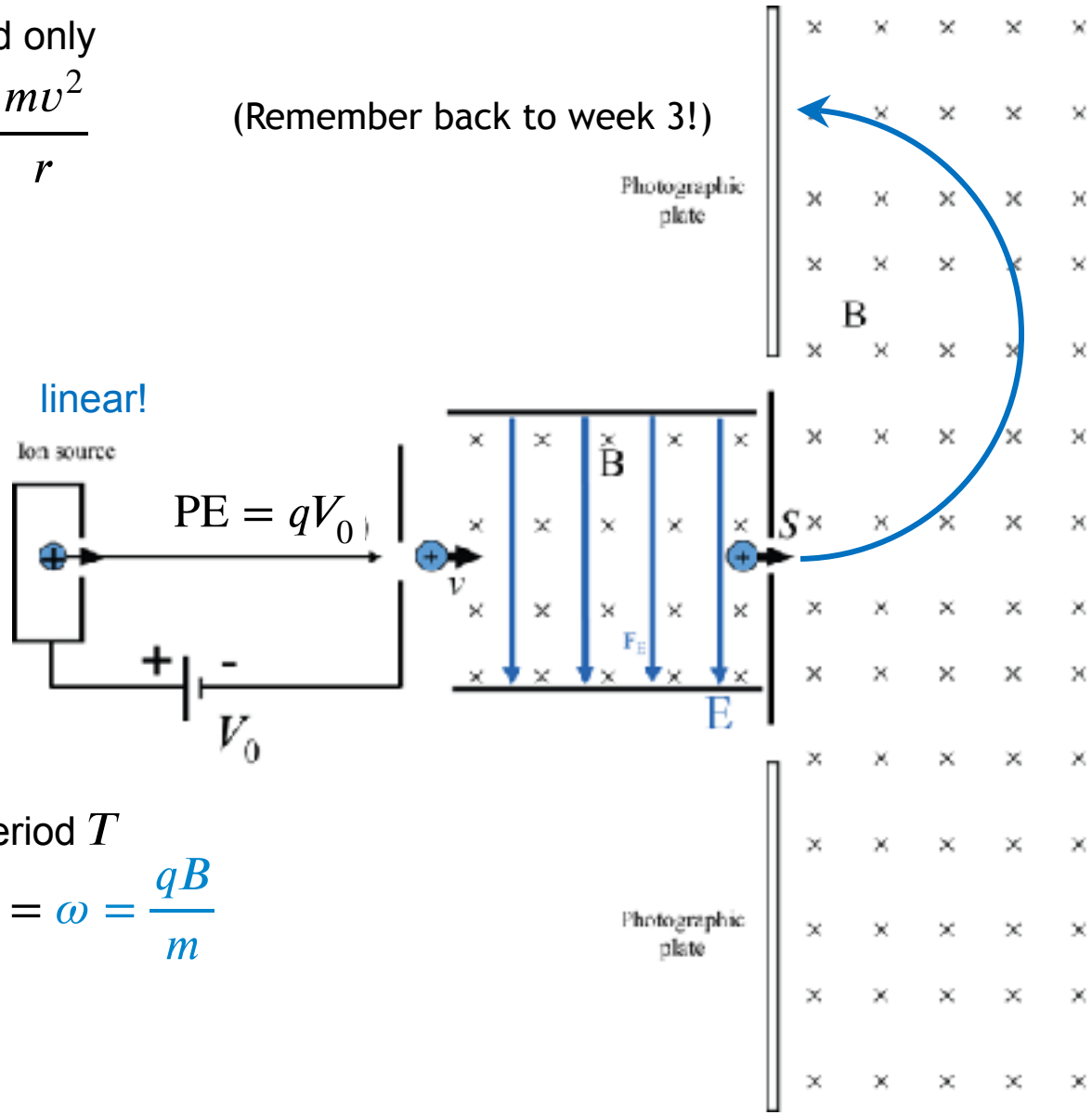

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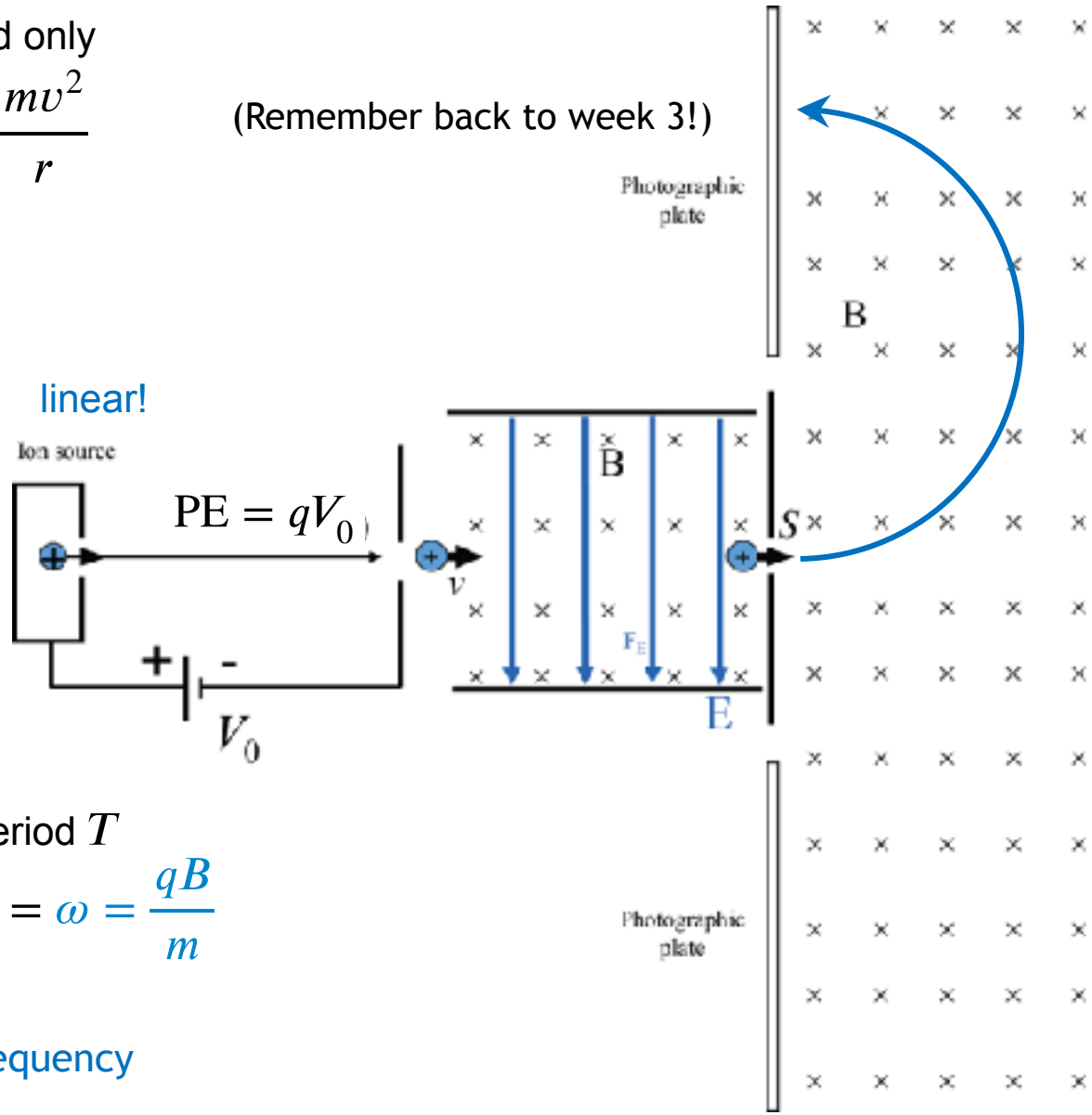

- Orbits: circumference $2\pi r$ in period T
  Velocity $v = \dfrac{2\pi r}{T}$ so $\dfrac{v}{r} = \dfrac{2\pi}{T} = \omega = \dfrac{qB}{m}$
- ω is known as the cyclotron frequency

Figure 1 is a cross-sectional view of the main features of an apparatus for measuring the mass to charge ratio of ions. The apparatus is enclosed in an airtight container (not shown) that has been evacuated.

A beam of positive ions from a discharge tube (not shown) is introduced into the apparatus and a thin beam of ions is formed by circular holes s_1 and s_2 . The thin beam passes between metal plates M and N. The plates are connected to a battery (not shown) which maintains a uniform electric field $\vec{E}$ (magnitude E) between them. Also in the region between the plates there is a uniform magnetic field $\vec{B}_1$ (magnitude B_1) which is perpendicular to the plane represented by the figure.

By suitably adjusting the magnitudes E and B_1 the thin beam can be made to pass undeflected between the plates M and N. Some of the ions in the thin beam subsequently pass through the narrow hole s_3 and enter a region of the apparatus in which there is a second uniform magnetic field $\vec{B}_2$ (magnitude B_2) perpendicular to the plane represented in the diagram. In this region, the ions are deflected by the magnetic field, so that they move along semicircular paths (diameter 2r) until they strike a photographic plate. The figure shows the impact points X and Y, of ions moving along two different semicircular paths.

Note: The formulas F = Q E and F= Q v B may be useful in answering the following questions:

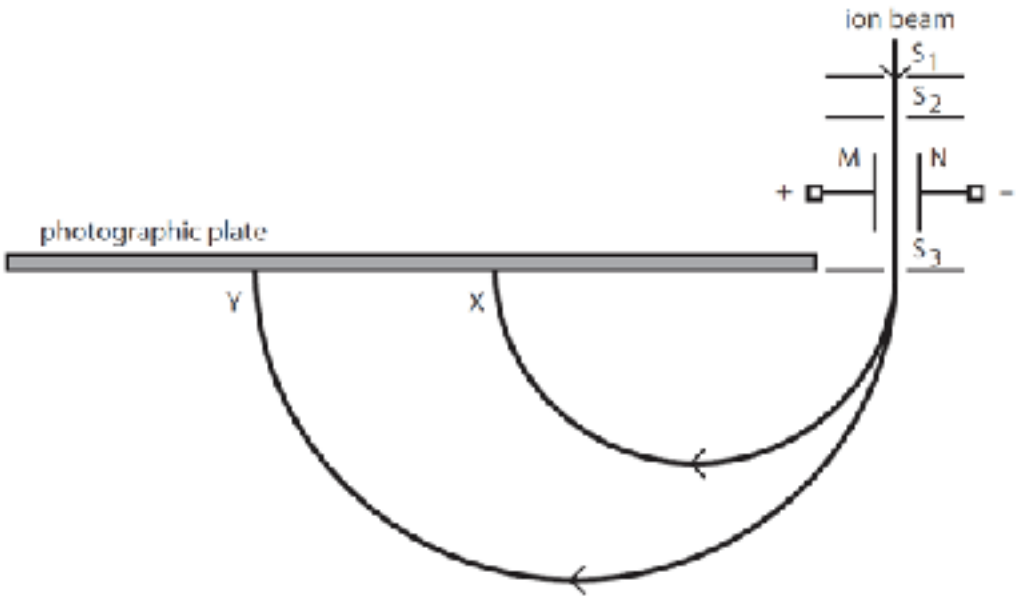

Figure 1

Mass Spectrometer - 1

41. Which of the following gives the speed of an ion passing through slit S3?

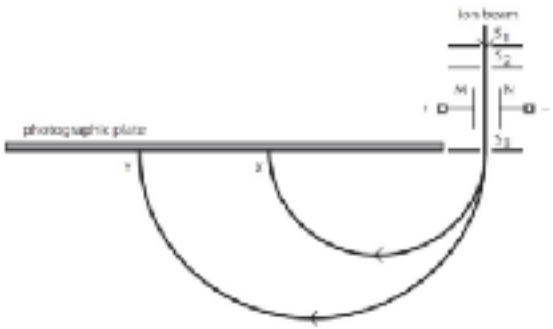

Figure 1

A. $\frac{B_1}{E}$

B. $\frac{E}{B_1}$

C. EB_1

D. $E - B_1$

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Mass Spectrometer - 2

41. The arrangement of electric and magnetic field in the region between plates M and is designed to:

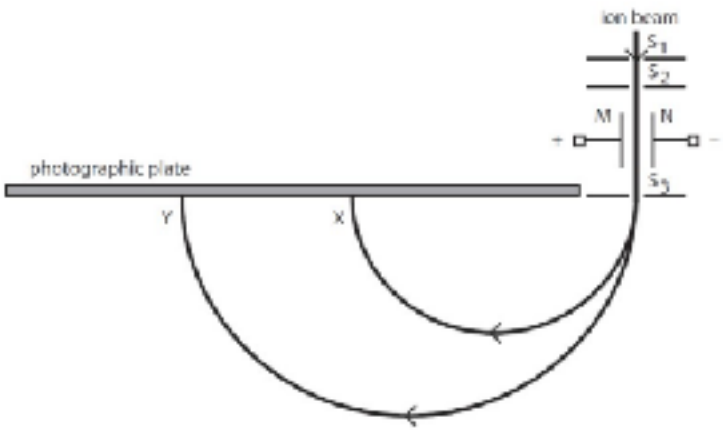

Figure 1

A. obtain a thin beam of ions at slit S3

B. reduce the number of ions arriving at S3

C. ensure that ions passing through S3 have the same velocity

D. ensure that ions through S3 have the same charge

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Mass spectrometer 3

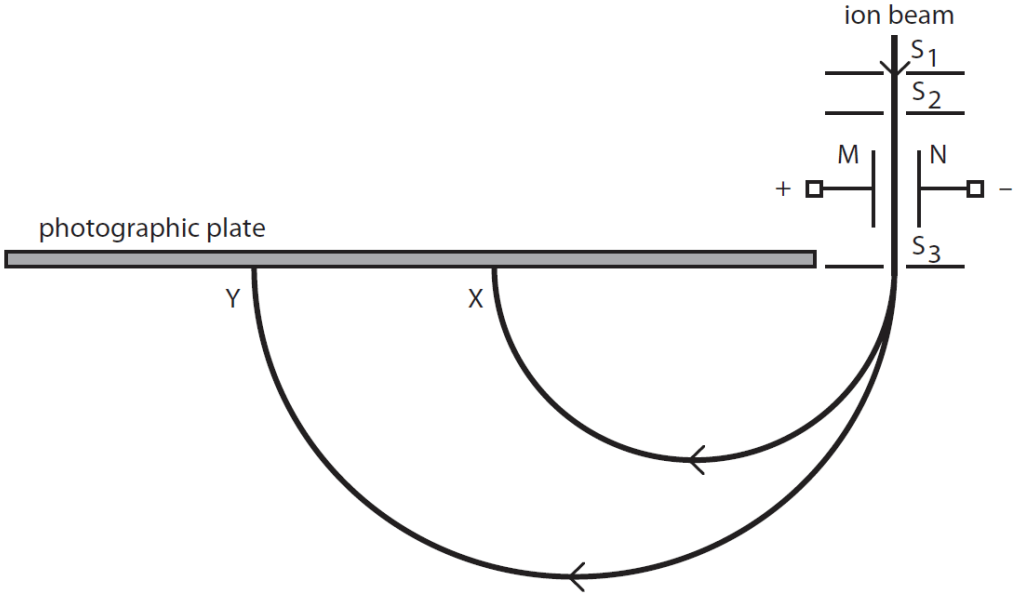
42 Which of the following gives the speed of an ion of mass m just before it strikes the photographic plate?

A $\dfrac{QrB_2}{m}$

B $\dfrac{m}{QrB_2}$

C $\dfrac{2QB_1}{m}$

D $\dfrac{m}{2QB_1}$



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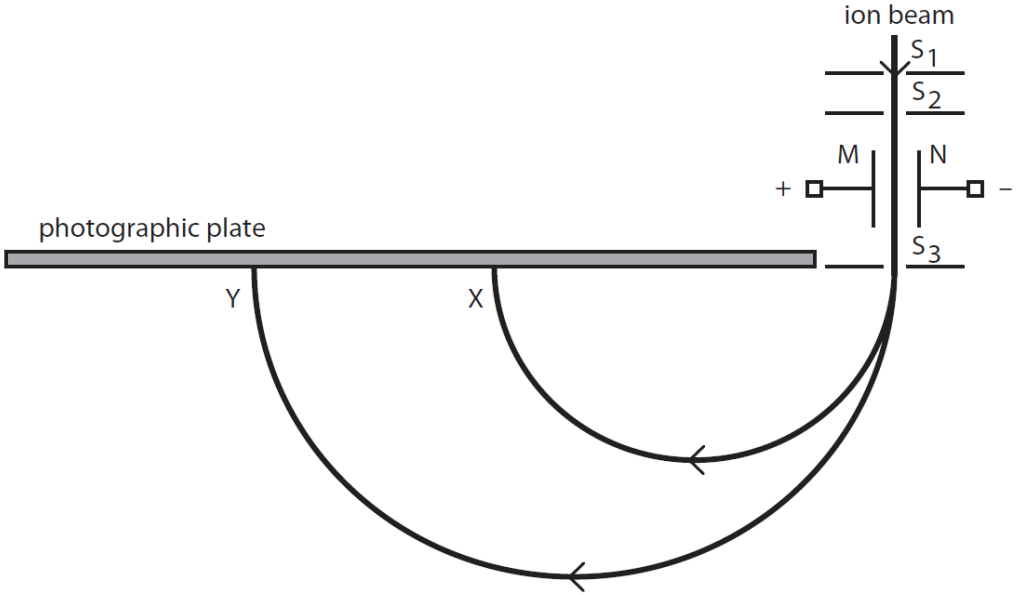

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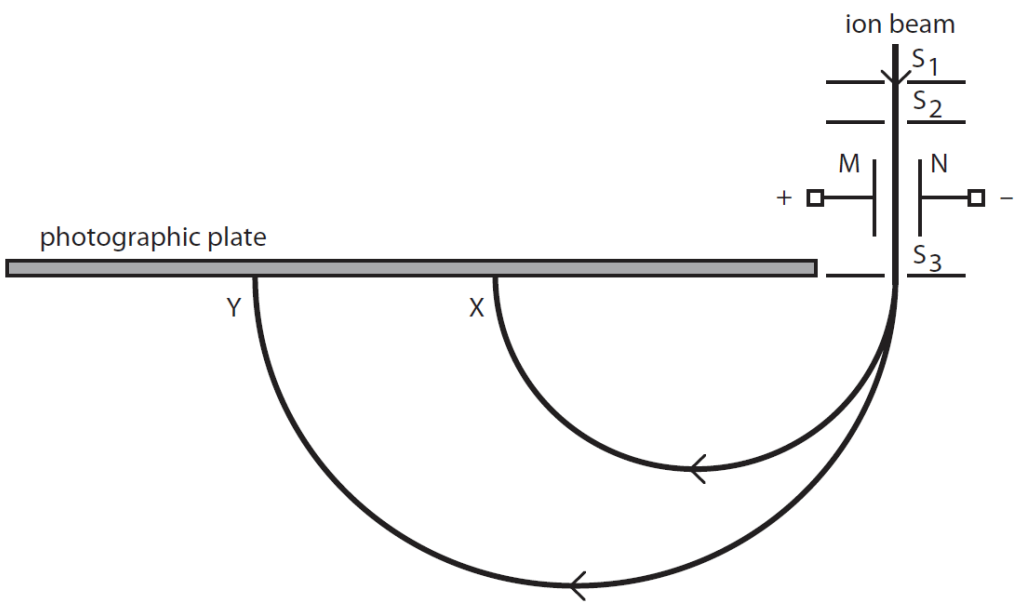
Equate magnetic, centripetal: $F = Qv \times B = QvB = \dfrac{mv^2}{r}$

Rearrange: $v = \dfrac{QrB}{m}$ where relevant B is B_2



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Mass spectrometer 4

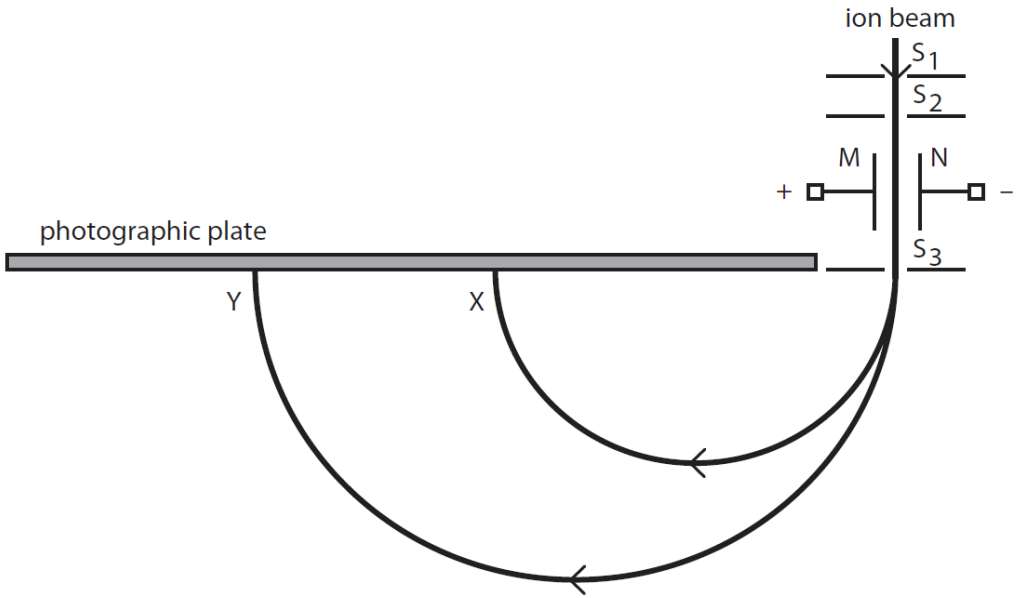
43 If the mass to charge ratios of negatively charged ions were to be determined by the apparatus, it would be simplest to reverse the direction of

A $\vec{E}$ only.

B $\vec{B}_2$ only.

C $\vec{E}$ and $\vec{B}_1$ only.

D $\vec{B}_1$ and $\vec{B}_2$ only.



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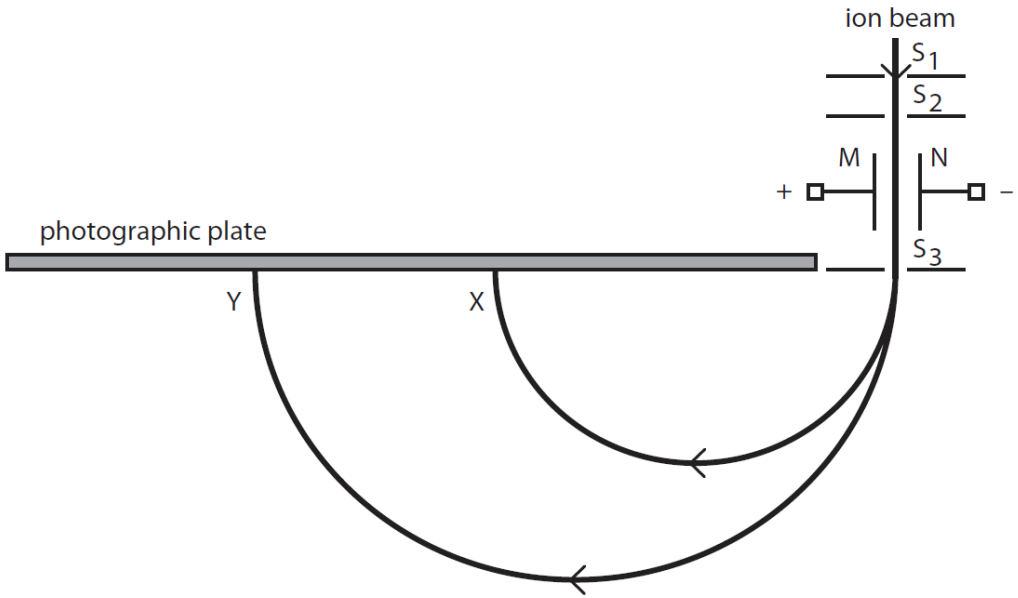

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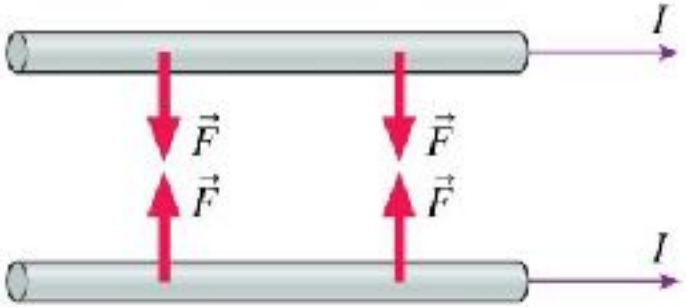
Magnetic forces on currents and charges

- Magnetic fields exert forces on moving charges, e.g. electric currents
- Since a current-carrying wire produces a magnetic field, there are magnetic forces between current-carrying wires



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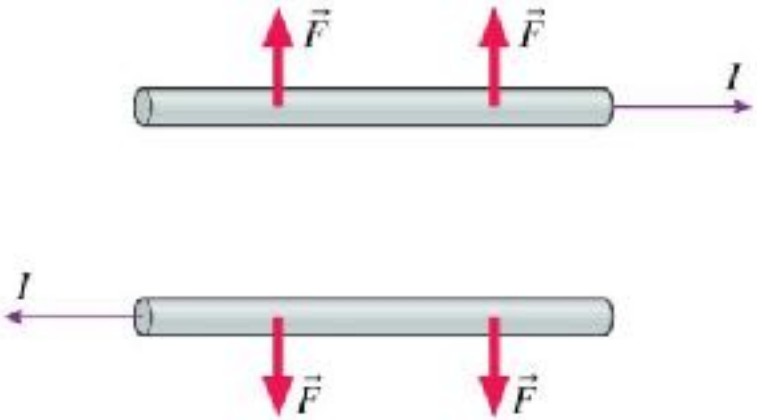

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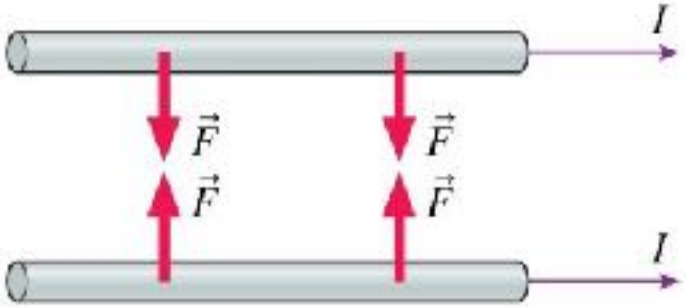

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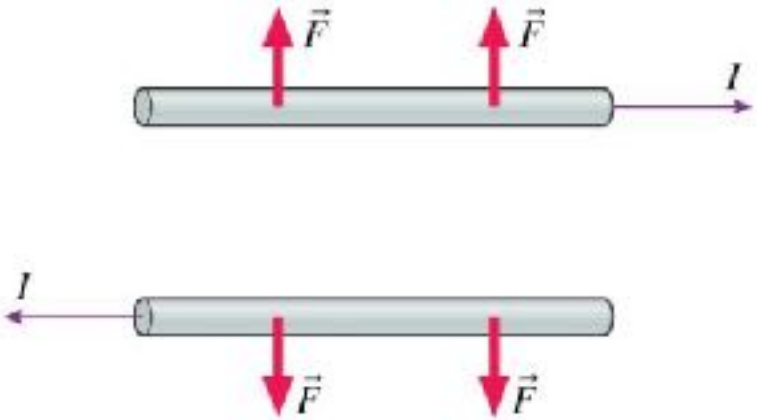

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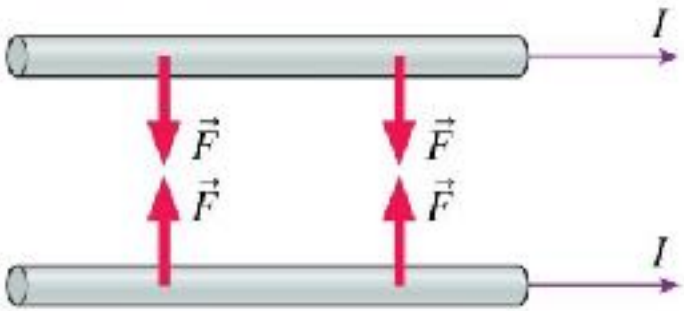

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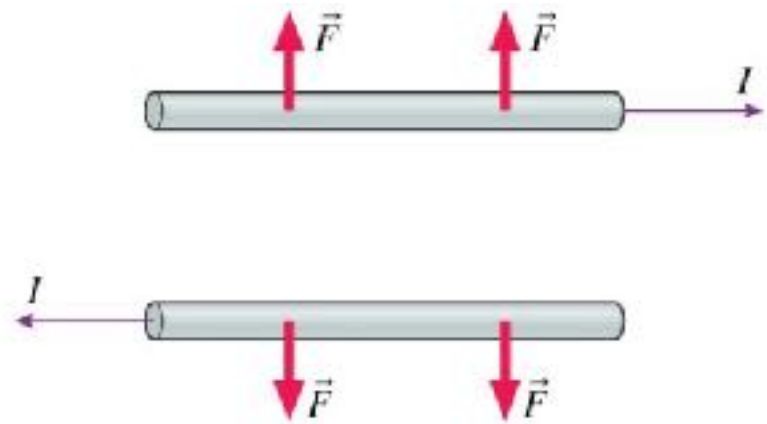

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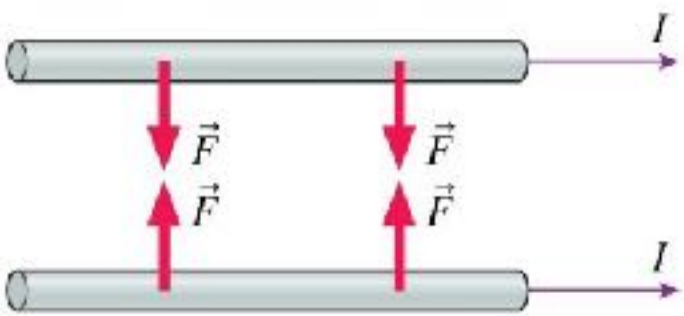

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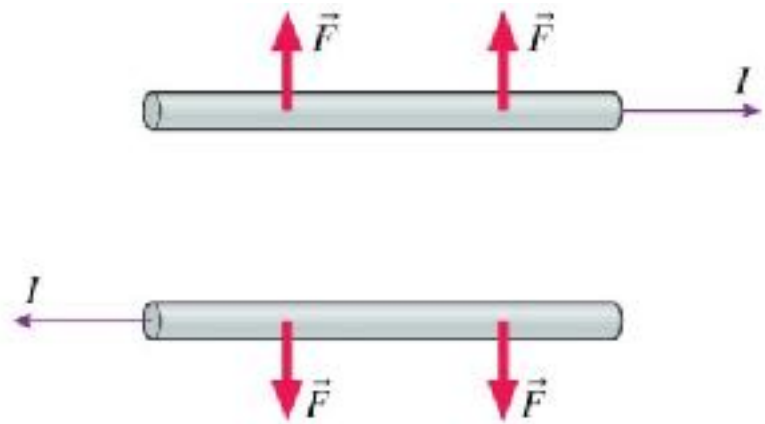

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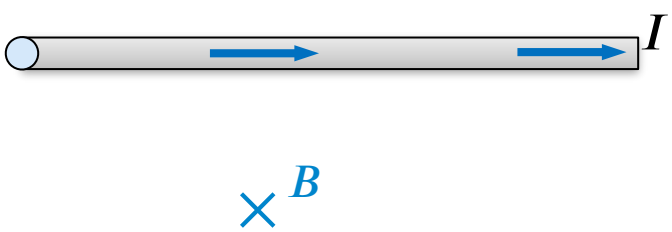

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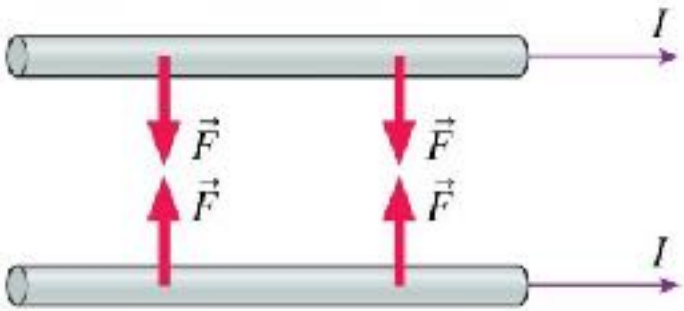

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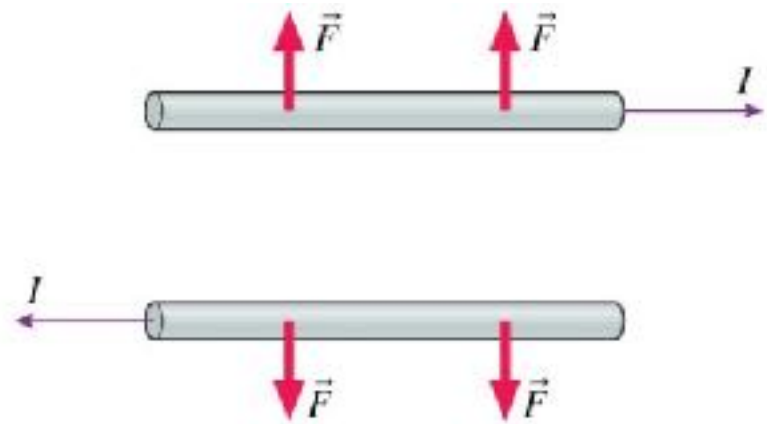

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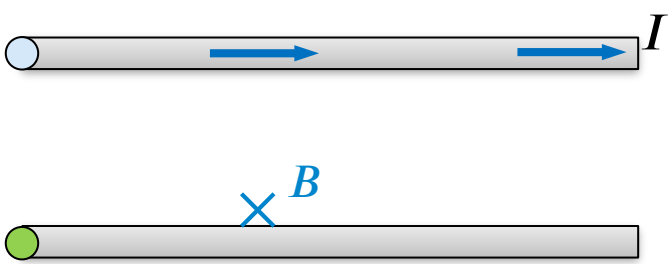

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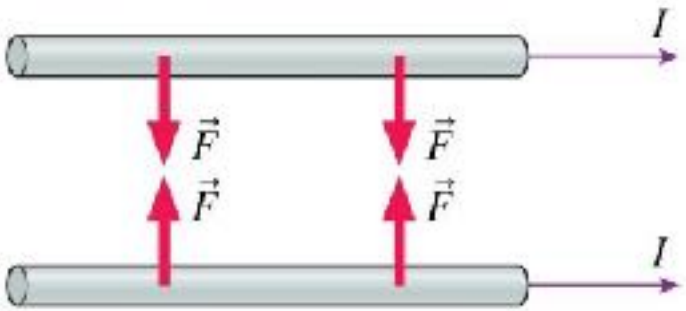

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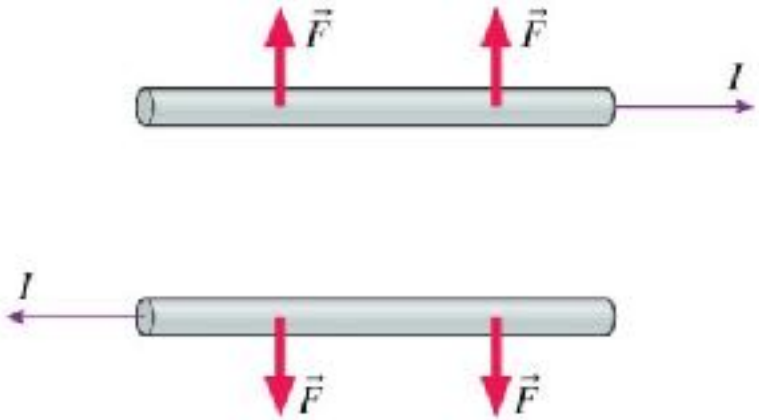

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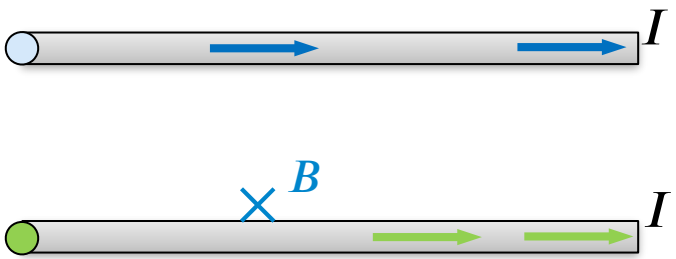

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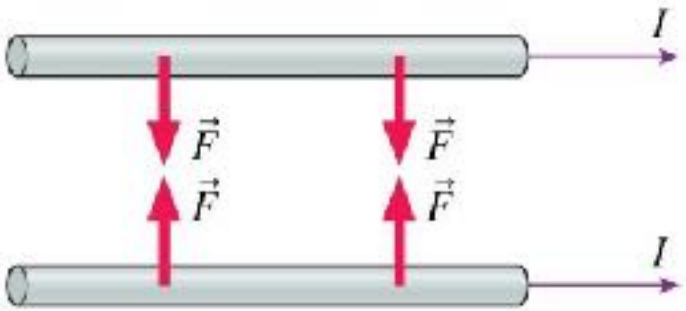

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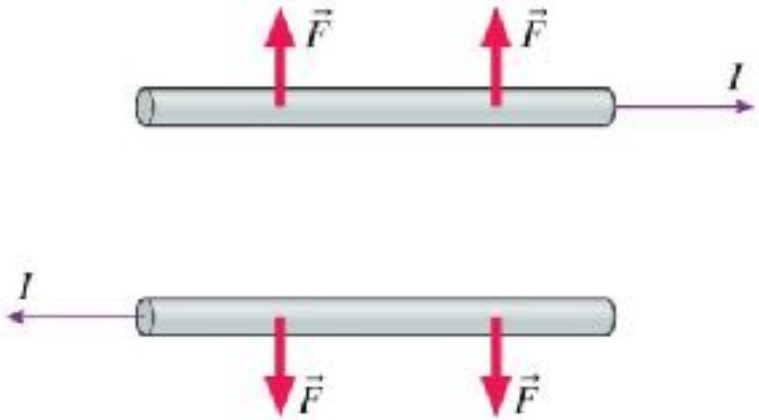

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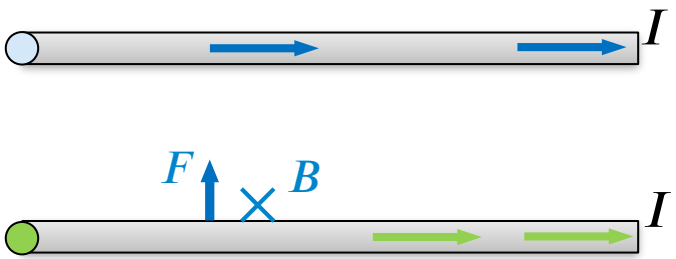

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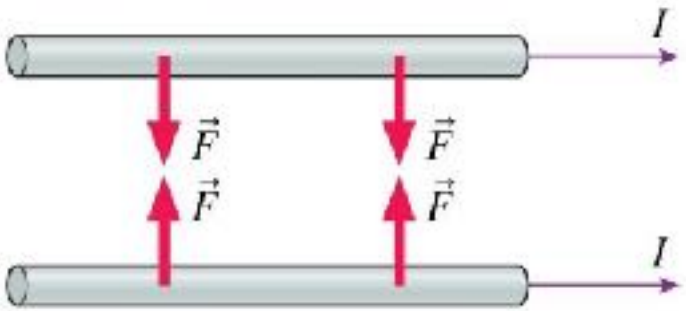
- Since a current-carrying wire produces a magnetic field, there are magnetic forces between current-carrying wires



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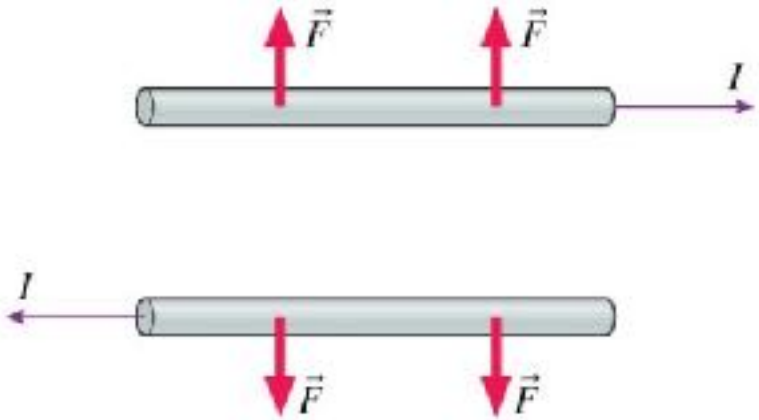

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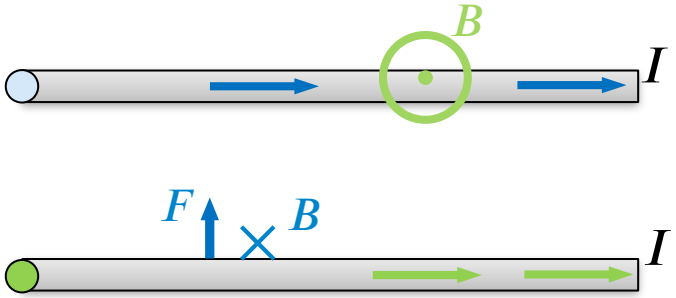

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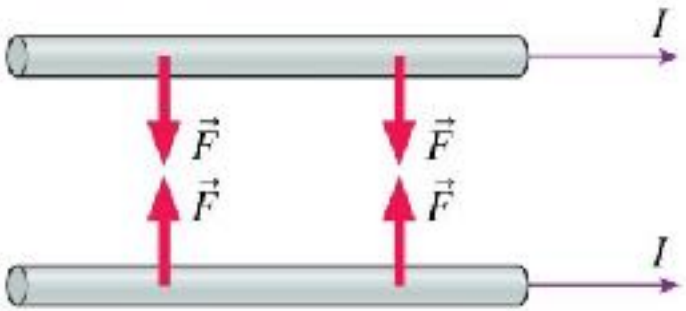

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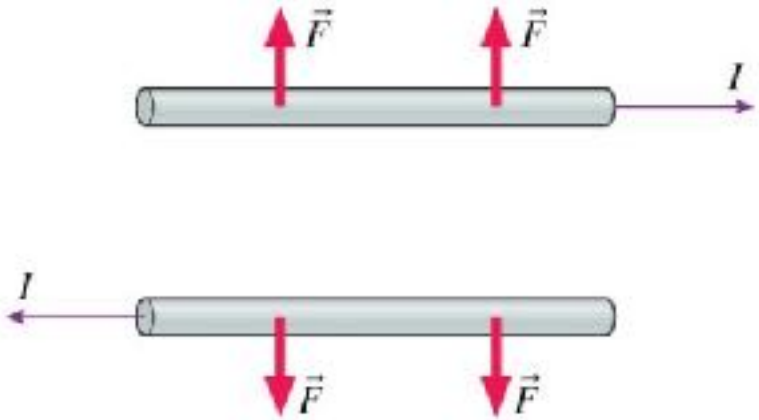

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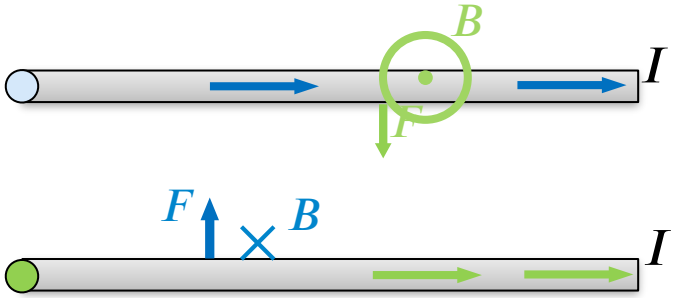

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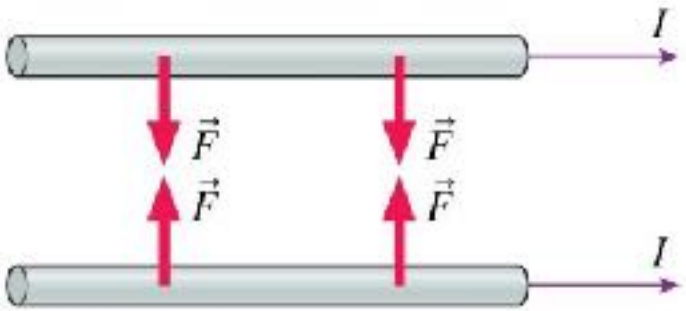

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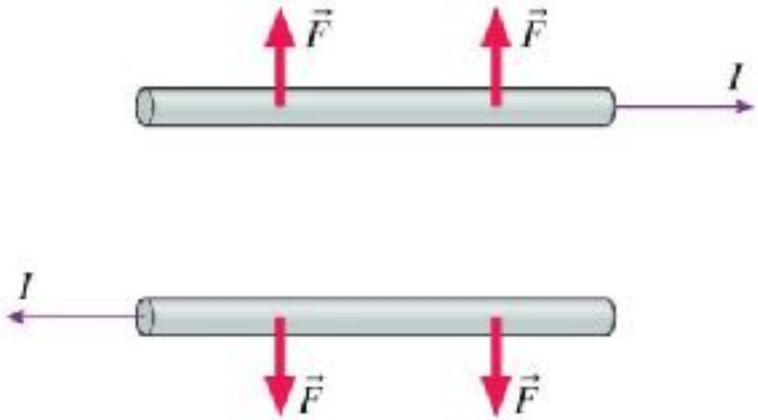

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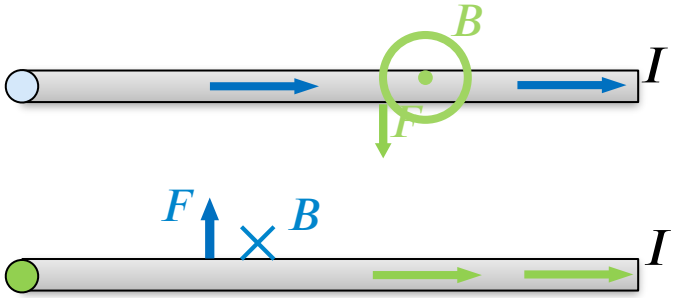


D Ea-1 Force on a Current Carrying Conductor in a magnetic field

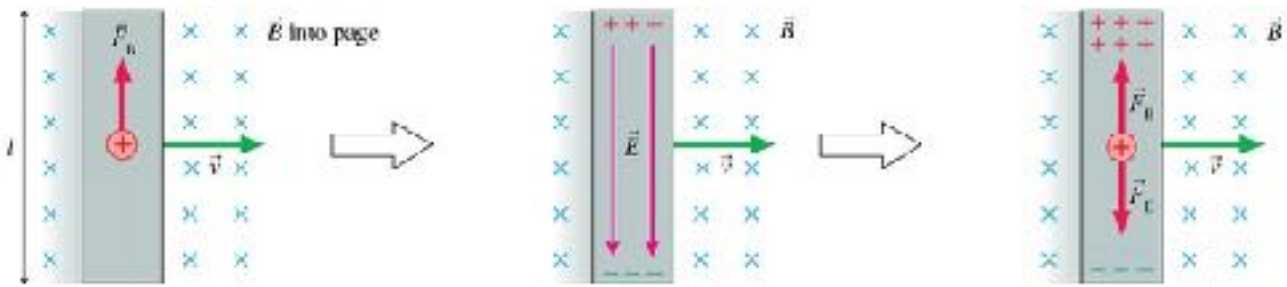
Chapter 21 Electromagnetic induction

Magnetism

- Induced Currents
- Magnetic Flux
- Faraday's Law of Induction
- Transformers

21.1 emf and electric current in wire

- Consider the charges inside the wire



If they're moving, they feel the usual magnetic force $F = qvB\sin\theta$

Charges move until $F = qE$ balances $F = qvB$

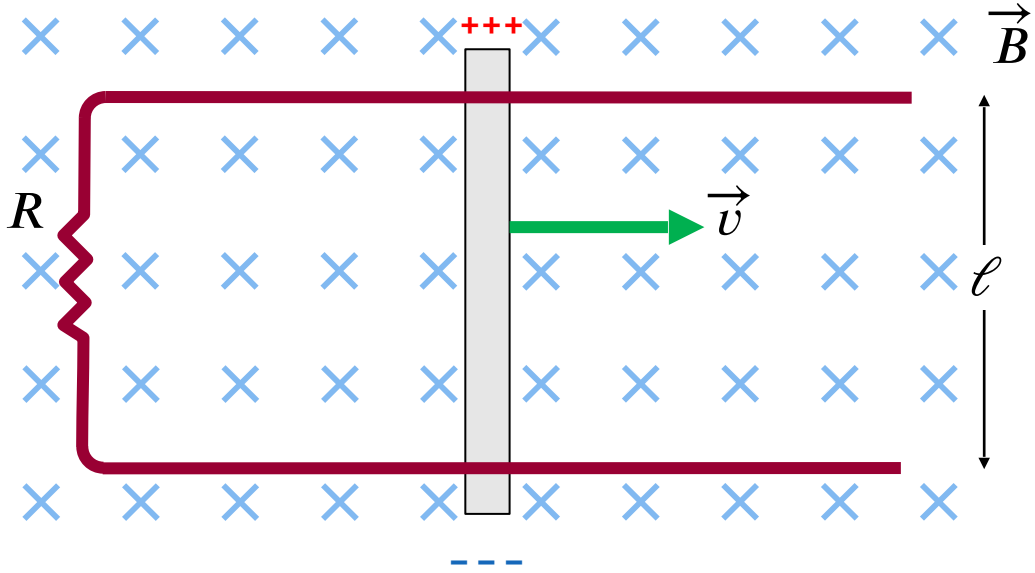
Electric field = change in Voltage divided by distance, so if Voltage (emf) is $\mathcal{E}$:

$$E = \frac{\mathcal{E}}{l} \Rightarrow q\frac{\mathcal{E}}{l} = qvB \Rightarrow \mathcal{E} = vlB$$

- Potential (emf) along wire given by $\varepsilon = v\ell B$

- Current depends on resistance in wire: $I = \frac{\varepsilon}{R} = \frac{v\ell B}{R}$

Example of how that emf can generate current



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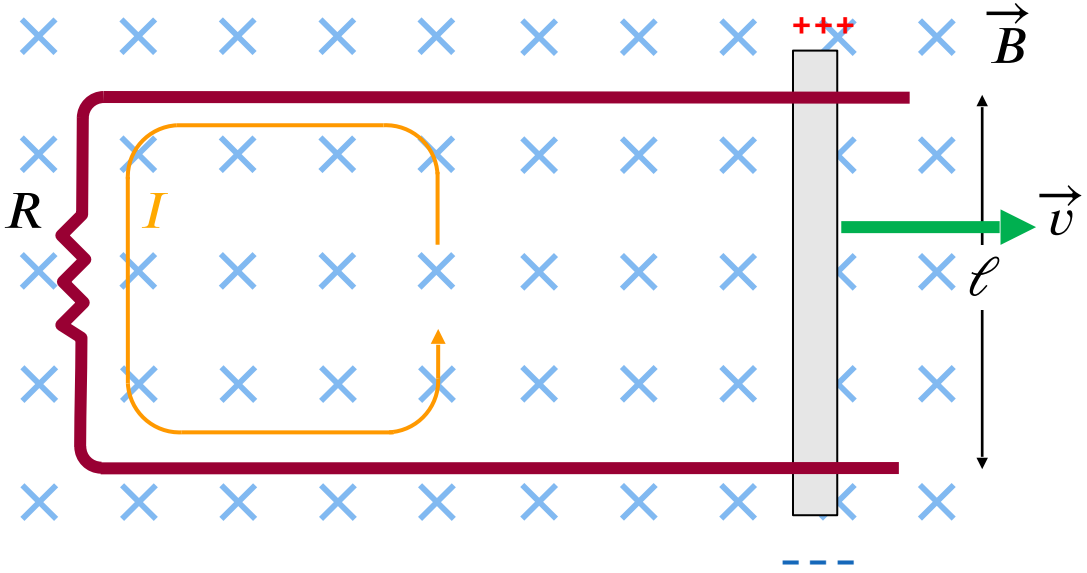
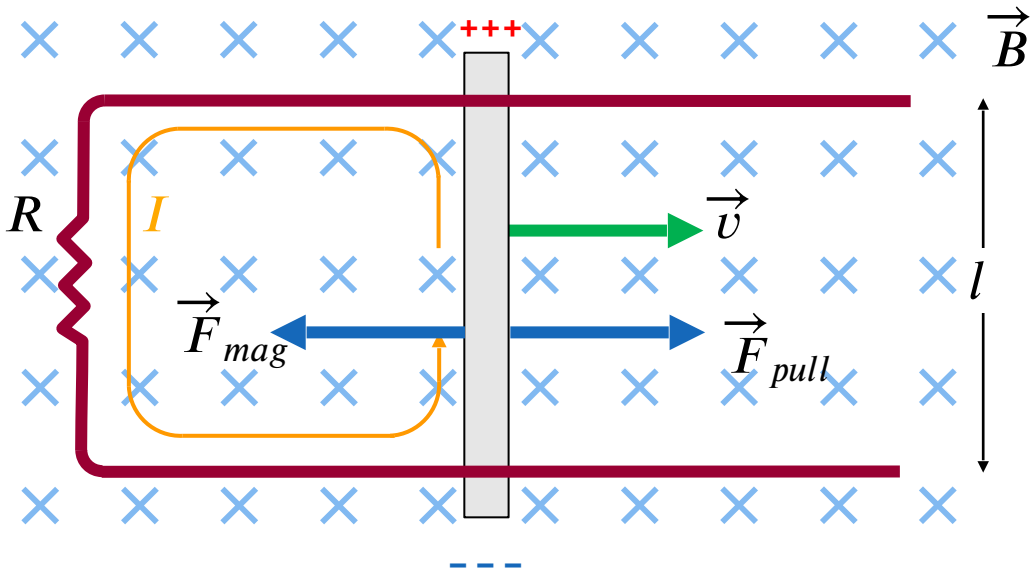


Induced current in a circuit



I is current in loop
$\vec{B}$ is magnetic field
R is resistance of loop
v is speed of wire
l is length of wire

- Current flows through the moving wire: $$I = \frac{\varepsilon}{R} = \frac{vlB}{R}$$

  (R is the resistance of the wire, or a resistor in the circuit)
- But there is a force on a current-carrying wire in a magnetic field: $\vec{F}_{mag} = IlB$
  (exercise: start with $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$)
- So we have to *pull* or *push* the wire to keep it moving at speed v

$$\vec{F}_{pull} = \vec{F}_{mag} = IlB = \left(\frac{vlB}{R}\right)lB = \frac{vl^2B^2}{R}$$

Energy and power

- Applying force F_{pull} to move wire so we must *do work*, P_{in} :

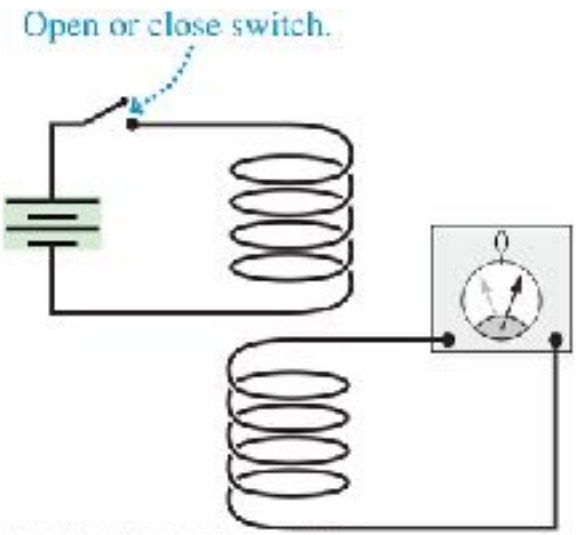
$$P_{in} = F_{pull} v = \frac{v^2 l^2 B^2}{R}$$

- Power dissipated P_{out} in heating of resistive elements in circuit:
  - Power out $P_{out} = I^2 R = \left(\frac{vlB}{R}\right)^2 R = \frac{v^2 l^2 B^2}{R}$
  - Same as power to pull the wire

Induced Currents

Demo: Eb-2

Induced Currents



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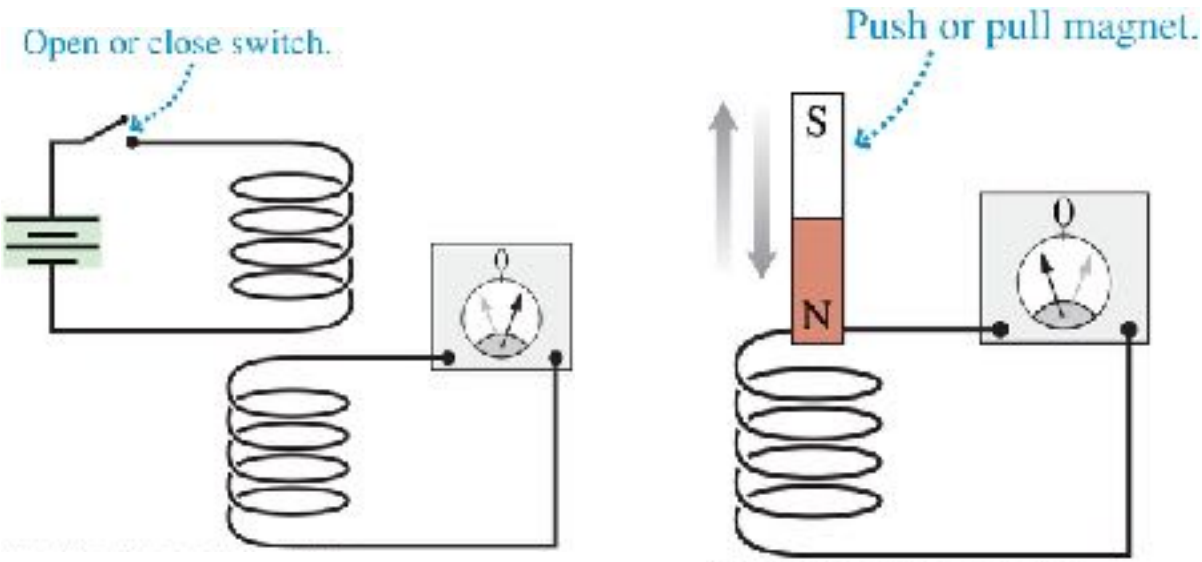

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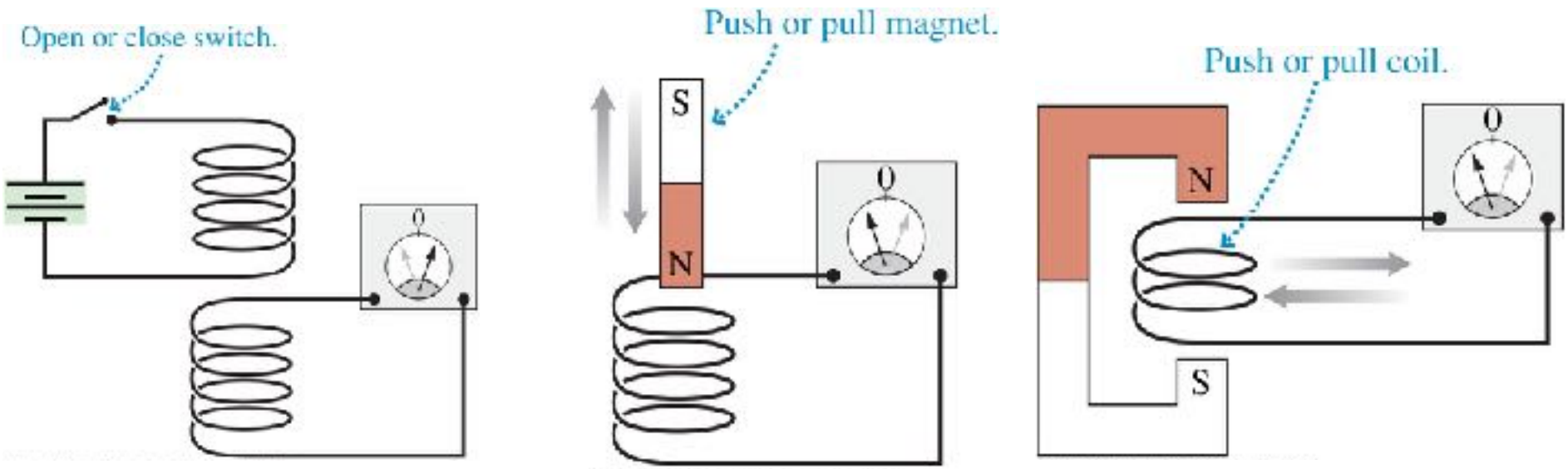
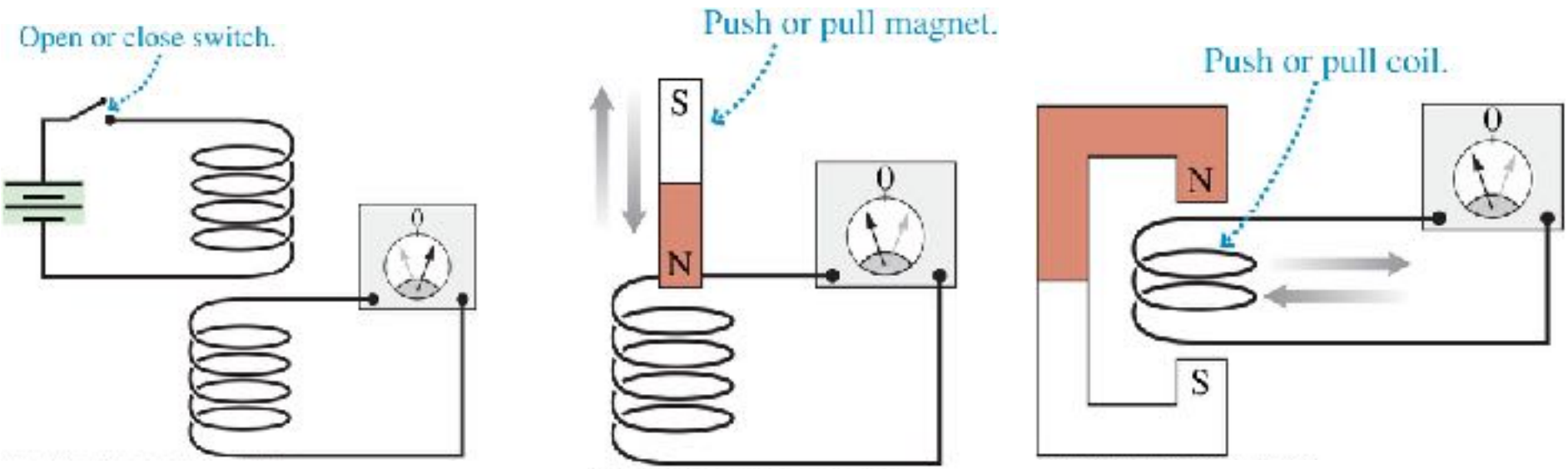


Demo: Eb-2

Induced Currents

- An electric field can produce a magnetic field (e.g. solenoid)
- Michael Faraday (1820's) demonstrated the reverse (*almost)*
- A ***changing*** magnetic field *induces* an electric current
- Current induced in a closed loop of wire that "surrounds" the changing magnetic field



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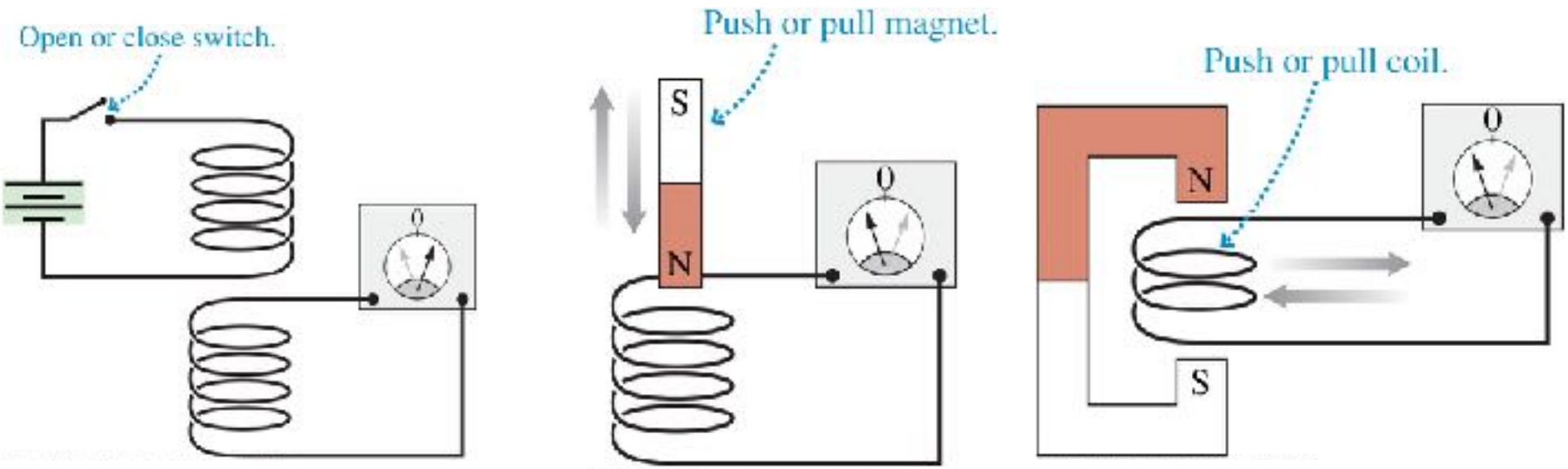

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D Eb-2 Faradays law of induction

Demo: Eb-2

Magnetic Flux

- Define magnetic flux as $\Phi = \vec{A} \cdot \vec{B} = AB\cos\theta$

$$\Phi = \vec{A} \cdot \vec{B}$$

$\vec{A}$ is vector area
- Direction normal to area
- Length proportional to area

$\vec{B}$ is magnetic field

θ is angle between $\vec{A}$ and $\vec{B}$

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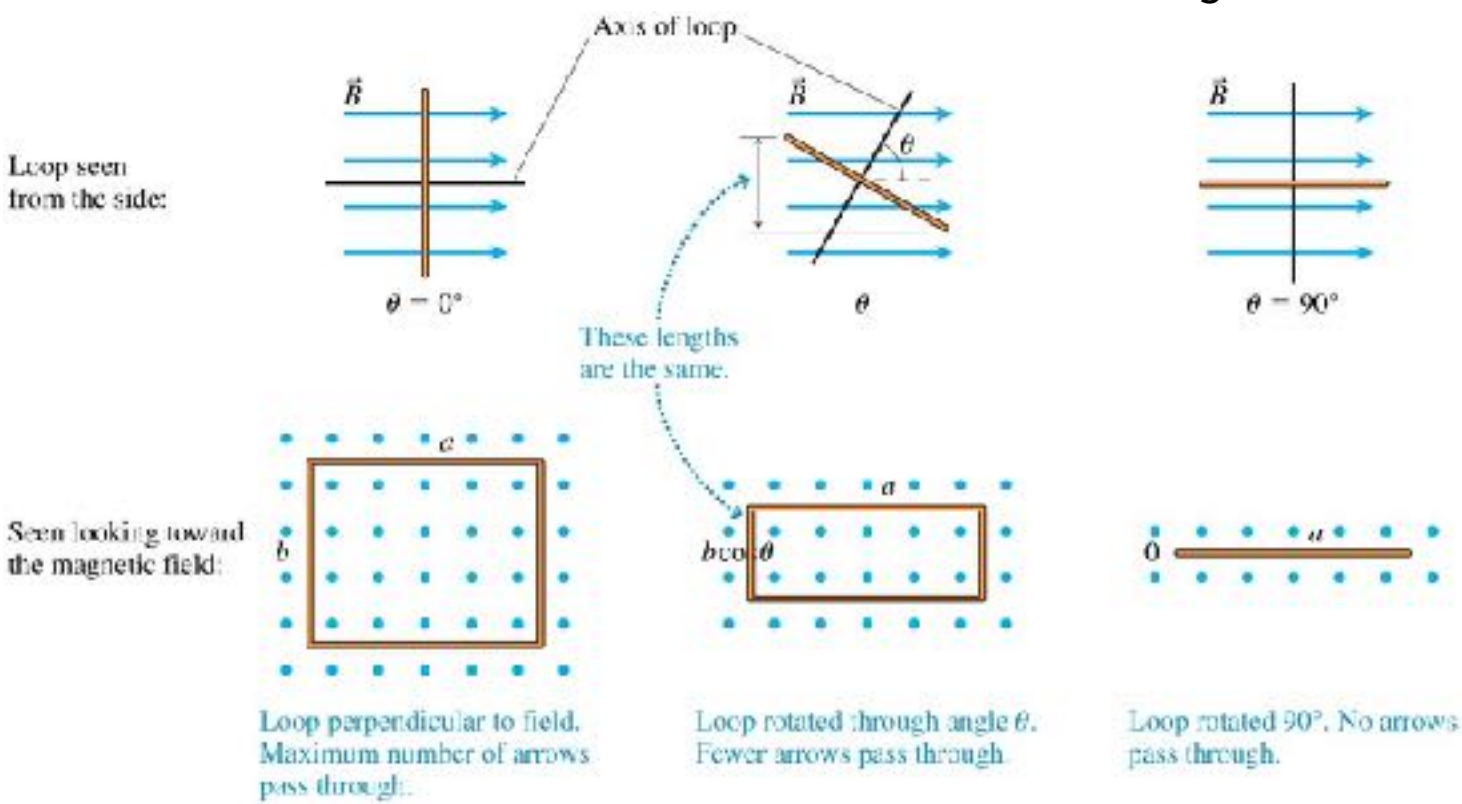

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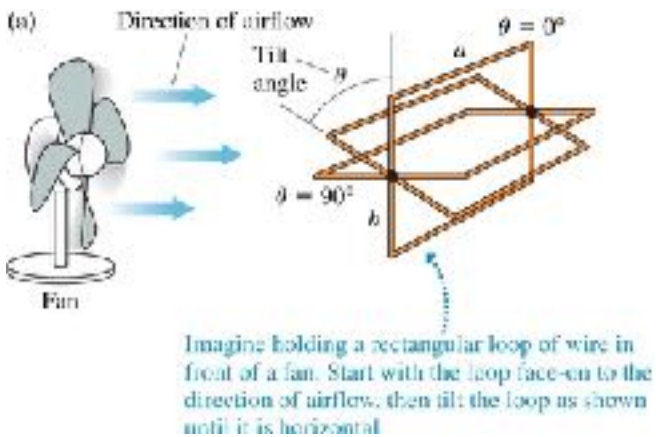

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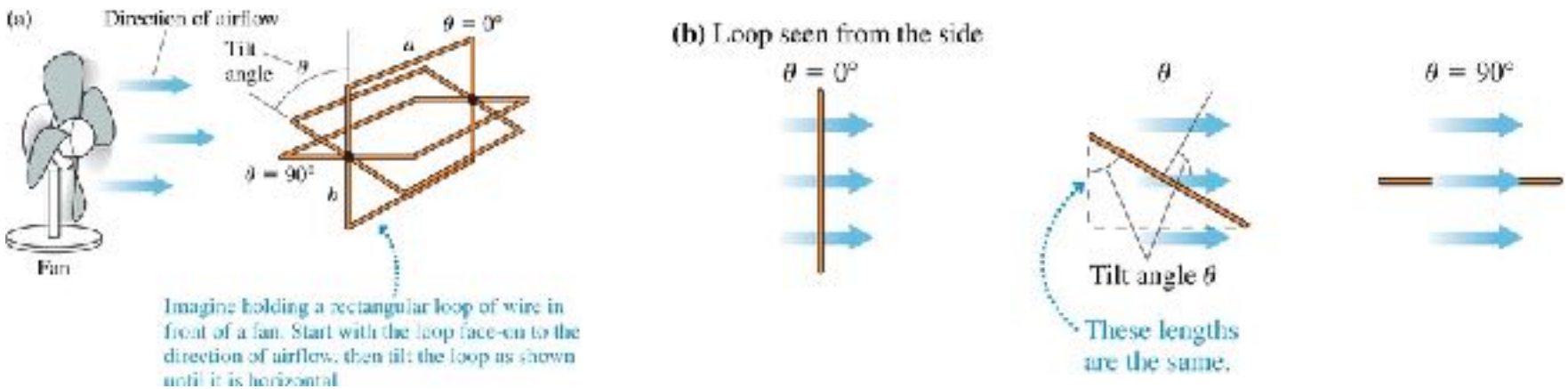

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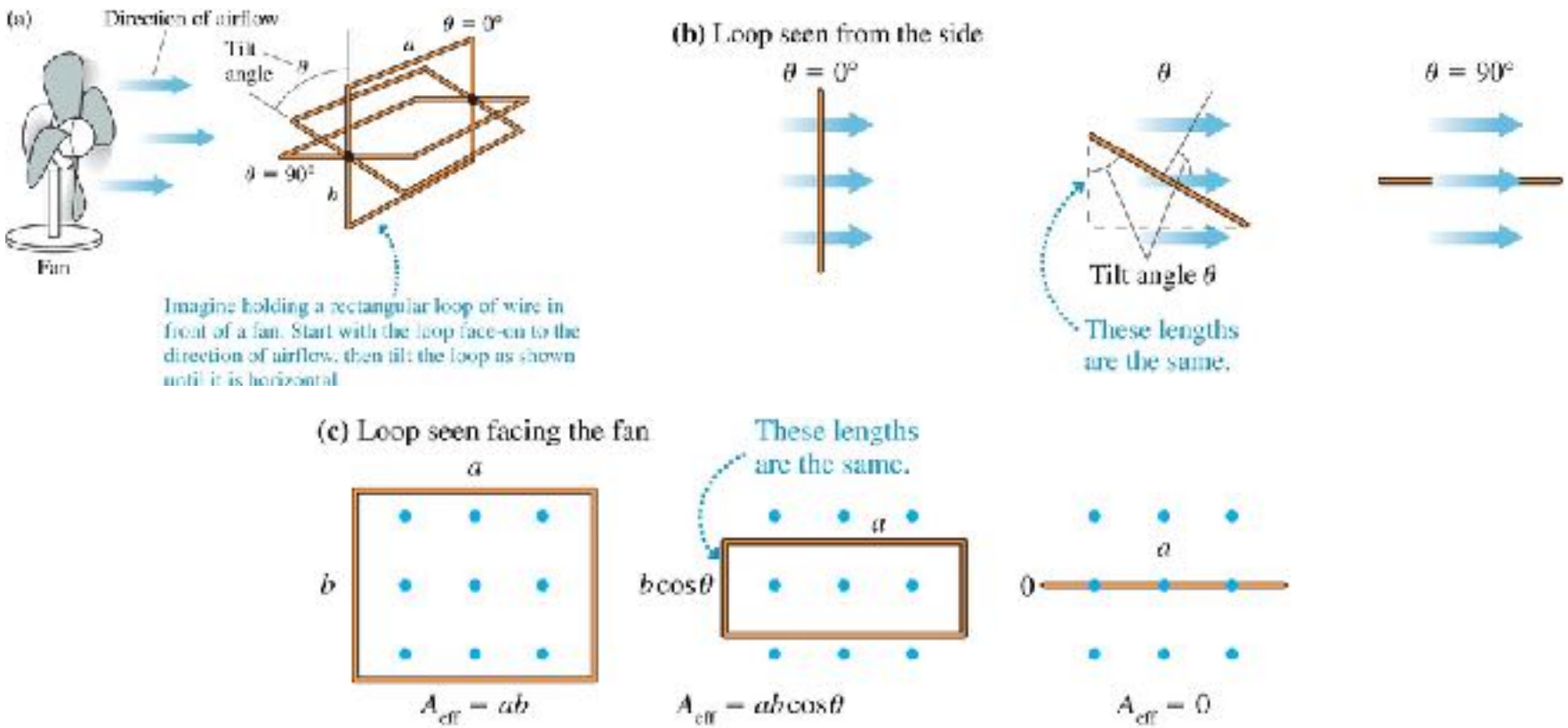

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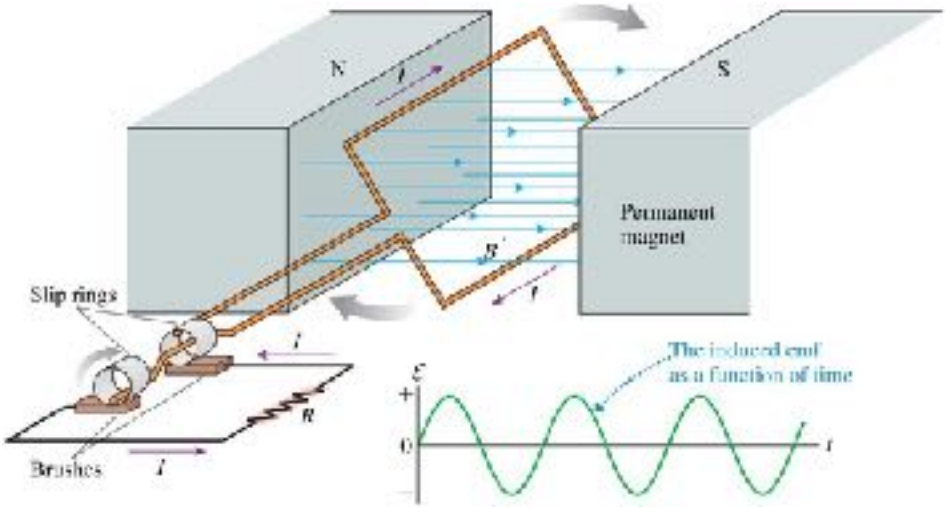

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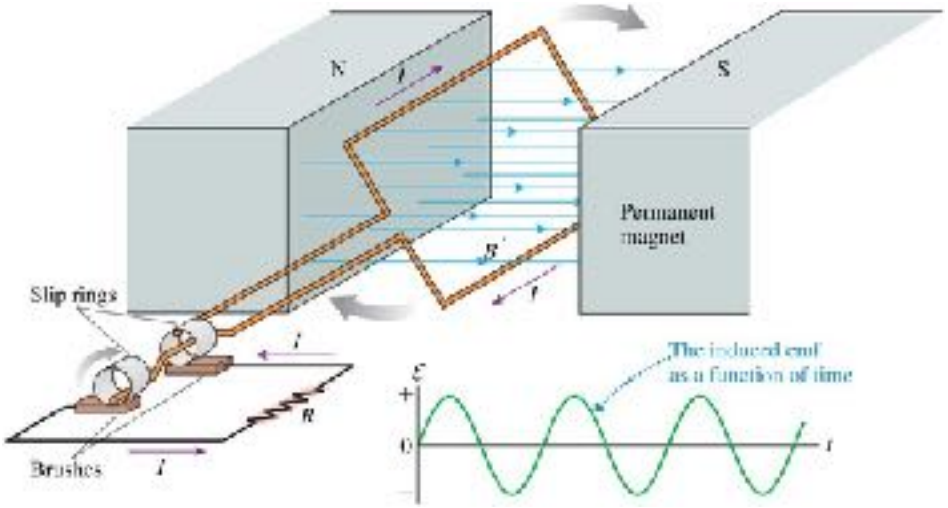
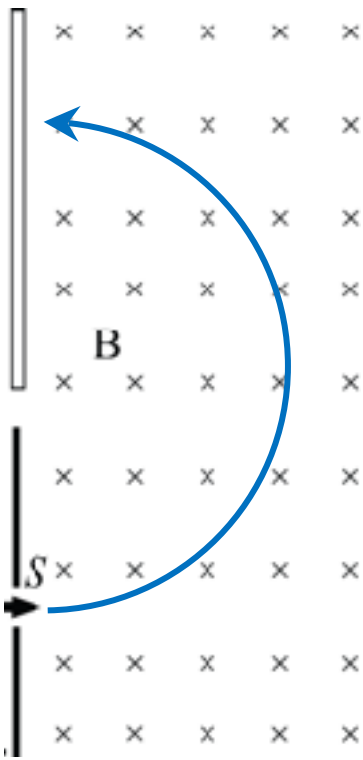


Summary

- Mass spectrometer
  - Additional B-only
  - Circular orbits
  - Balance centripetal and qvB



- Changing magnetic field induces current
- Define magnetic flux as $\Phi = \vec{A} \cdot \vec{B} = AB\cos\theta$
- Convert mechanical power into electrical power